

Session 4 :

Types of Machine Learning

Now that you know what Machine Learning is, it's time to discover the different ways machines can learn. Just like you have different ways of learning - sometimes from a teacher, sometimes by exploring on your own, sometimes by trial and error - machines have their own learning styles too.

Get ready!



Recap : What We Learned in Session 3

Last time, we discovered what Machine Learning is. Let's refresh our memory:

Concept	Key Takeaway
Machine Learning	Computers learning patterns from examples instead of following fixed rules
Traditional Programming	Writing step-by-step instructions for a computer
Rule-Based Systems	Programs that follow fixed rules - great for simple tasks, fail for complex ones
Pattern Recognition	Finding regularities in data - the foundation of ML
Learning from Patterns	Using found patterns to make predictions on new data
ML in Everyday Life	TikTok, Spotify, Netflix, Google Maps - all use ML

Today we'll explore the different ways machines can learn. Think of it like discovering that there's not just one way to learn

Objectives

By the end of this session, you will be able to:

01

Identify Learning Styles

Understand that machines can learn in different ways

02

Define Supervised Learning

Explain learning with labeled examples

03

Define Unsupervised Learning

Explain finding patterns without labels

04

Define Reinforcement Learning

Explain learning from rewards and penalties

05

Give Real Examples

Match each ML type to real-world applications

06

Compare Approaches

Tell the difference between the three types

07

See the Big Picture

Understand that real AI systems often combine multiple types



Big Question of the Day

How many different ways can machines learn?

Think about how **YOU** learn:

- Sometimes you have a teacher giving you the answers (like flashcards)
- Sometimes you explore and figure things out on your own
- Sometimes you learn by trying things and seeing what works

📄 Machines have the same **three main ways** to learn! By the end of this session, you'll know all three and be able to spot them in the real world.

Different Ways Machines Learn

Just like humans have different learning styles, machines have different ways of learning too. The type of learning depends on:



What data we have

Labeled? Unlabeled? Interactive?



What we want the machine to do

Predict? Find groups? Make decisions?



How we give feedback

Correct answers? Patterns?
Rewards?

The Three Main Types of Machine Learning



Supervised Learning

Learning Style: Learning with a teacher

Analogy: Studying with flashcards that have answers on the back



Unsupervised Learning

Learning Style: Learning by exploring

Analogy: Organizing a messy closet without instructions



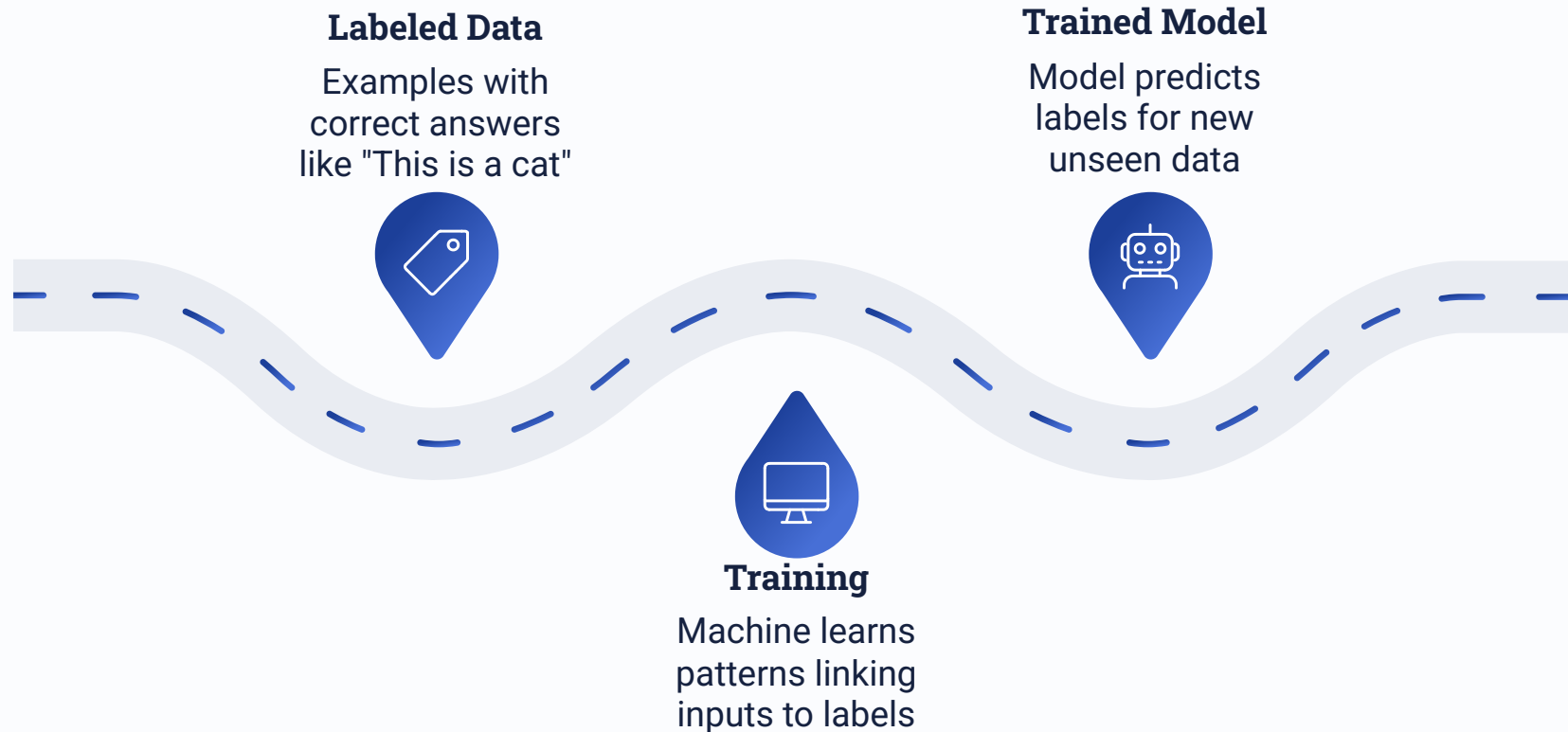
Reinforcement Learning

Learning Style: Learning by trial and error

Analogy: Playing a video game and learning from points

Supervised Learning - Learning with Labels

Supervised Learning is when we teach a machine using **labeled examples**. Each example comes with the correct answer, like flashcards with the answer on the back.



The machine receives labeled data, learns patterns during training, and then can predict labels for new, unseen data.

Supervised Learning - Two Important Tasks

Supervised learning can be used for two main kinds of problems:

Task 1: Classification - Sorting Things into Categories

Putting things into categories - The AI predicts which category or class something belongs to.

Examples	What's Being Predicted
Spam filter	"Spam" or "Not Spam"
Face recognition	"This is Sara" or "This is not Sara"
Cat vs Dog classifier	"Cat" or "Dog"

📌 **Key Idea:** The answer is a category - there are a fixed number of possible answers.

Classification - Sorting Things into Categories

Classification is a type of Supervised Learning where the AI's job is to put something into a category.

How It Works

Training

The AI is trained on labeled examples (each example already has the correct category).

Predicting

When it sees new data, it predicts which category it belongs to.

- ❏ Imagine you have three bins: **"Toy," "Book," and "Game."** A friend gives you an object and asks, "Which bin does this go in?" That's classification – you're picking from the three bins.



Supervised Learning

Real-World Examples You Know

Classification Examples



Email Spam Filter

Classifies emails as "spam" or "not spam"



Google Photos Search

Classifies photos as "beach," "mountain," "birthday party"



Face Recognition

Classifies which person is in the photo



Snapchat Filters

Classifies where your eyes, nose, and mouth are

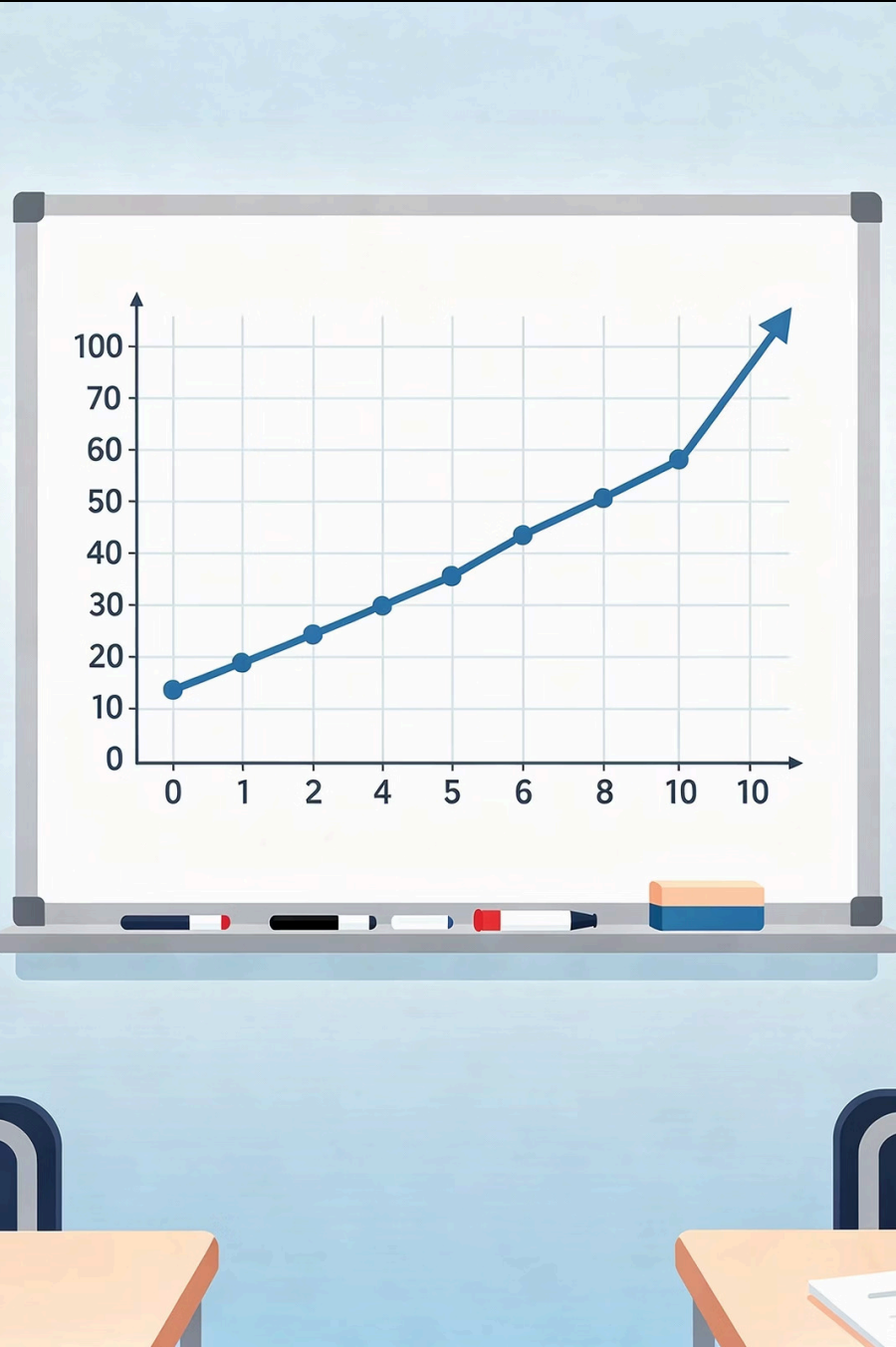
Task 2: Regression

Predicting numbers - The AI predicts a number or value.

Examples	What's Being Predicted
Weather app	Tomorrow's temperature (72°F, 68°F, etc.)
Sports scores	How many points a team will score
Car value	How much a used car is worth

 **Key Idea:** The answer is a number - there are infinite possibilities!

Both classification and regression use labeled data - they're both **Supervised Learning!**



Regression - Predicting a Number

Regression is another type of Supervised Learning where the AI's job is to predict a number. Think of it like a fill-in-the-blank: the answer can be any number (within a range), not just a few choices.

How It Works

Training on Labeled Examples

The AI is trained on labeled examples where each example has a number as the answer.

Predicting a Number

When it sees new data, it predicts a number that fits the pattern.

- ❏ Imagine you have a chart showing how many people came to school events and how many pizzas were ordered. Someone says, "Next event, we expect 50 people. How many pizzas should we order?" You use the pattern to estimate a number - that's regression.

Regression Examples



Weather App

Predicts tomorrow's temperature



Real Estate Apps

Predicts house prices



Google Maps

Predicts travel time to your destination



Uber

Predicts ride fare before you book

Classification vs. Regression

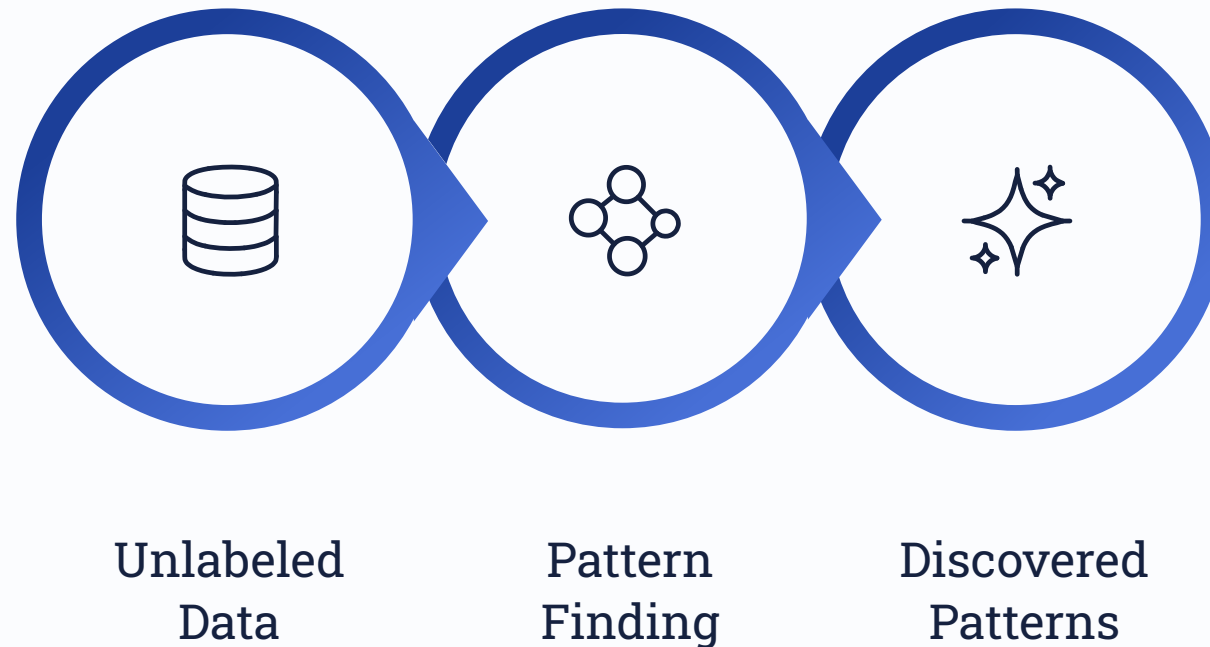
Both are Supervised Learning, both use labeled data, both learn from examples, but they answer different types of questions.

Classification	Regression
"Which category?"	"How many?" or "How much?"
A category (from a fixed list)	A number (any value)
Multiple-choice test	Fill-in-the-blank
"Is this movie suitable for kids, teens, or adults?"	"How many tickets will be sold for the movie?"
"What breed is this dog?" (husky, beagle, poodle)	"How much will this puppy weigh when fully grown?"
"Will it rain today?" (yes/no)	"How many inches of rain will fall?"

 When building an AI system, the first question is: **"What do we want to predict?"** If the answer is a category, we use **classification**. If it's a number, we use **regression**. Choosing the right approach makes the AI work correctly.

Unsupervised Learning - Finding Hidden Patterns

Unsupervised Learning is when we give a machine **unlabeled data** and let it find patterns on its own. There's no teacher telling it the right answers - it has to discover structure by itself.



The machine receives unlabeled data, searches for similarities, and discovers hidden groups or structures on its own.

Real-World Examples You Know

Clustering Examples



Google News

Groups news articles about the same topic together - without anyone labeling them



Customer Segmentation

Finds groups of shoppers with similar buying habits (budget shoppers, luxury shoppers, etc.)



Spotify / Apple Music

Groups songs with similar sounds or styles to create playlists



Movie Recommendations

Groups movies that similar types of people enjoy

Mini Activity 1 : Supervised or Unsupervised?

For each scenario, identify: Which type of ML? (Supervised, Unsupervised, or Reinforcement). If Supervised: Is it Classification or Regression?

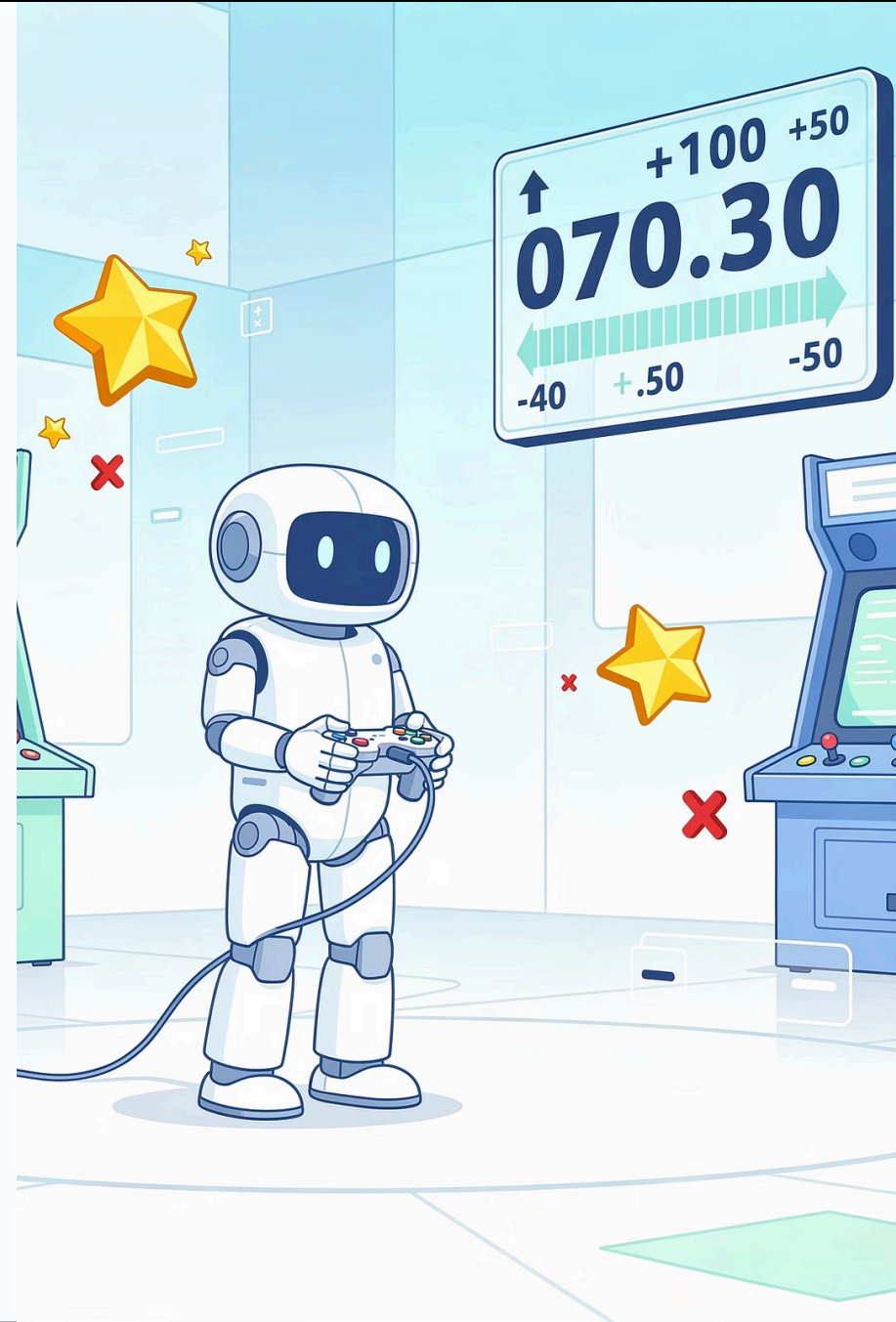
Scenario	ML Type	If Supervised: Class or Reg?
Predicting tomorrow's temperature		
Grouping customers by shopping habits		
An AI learning to play chess		
Identifying if an email is spam		
Predicting house prices		
Google News grouping articles		
Recognizing if a photo shows a cat		
A robot learning to walk		
Predicting how many points a team will score		
Spotify grouping similar songs		

Mini Activity 1 : Answers

Scenario	ML Type	If Supervised: Class or Reg?
Predicting tomorrow's temperature	Supervised	Regression (predicting a number)
Grouping customers by shopping habits	Unsupervised (clustering)	–
An AI learning to play chess	Reinforcement	–
Identifying if an email is spam	Supervised	Classification (spam or not)
Predicting house prices	Supervised	Regression (predicting a number)
Google News grouping articles	Unsupervised (clustering)	–
Recognizing if a photo shows a cat	Supervised	Classification (cat or not)
A robot learning to walk	Reinforcement	–
Predicting how many points a team will score	Supervised	Regression (predicting a number)
Spotify grouping similar songs	Unsupervised (clustering)	–

Reinforcement Learning - Learning from Rewards

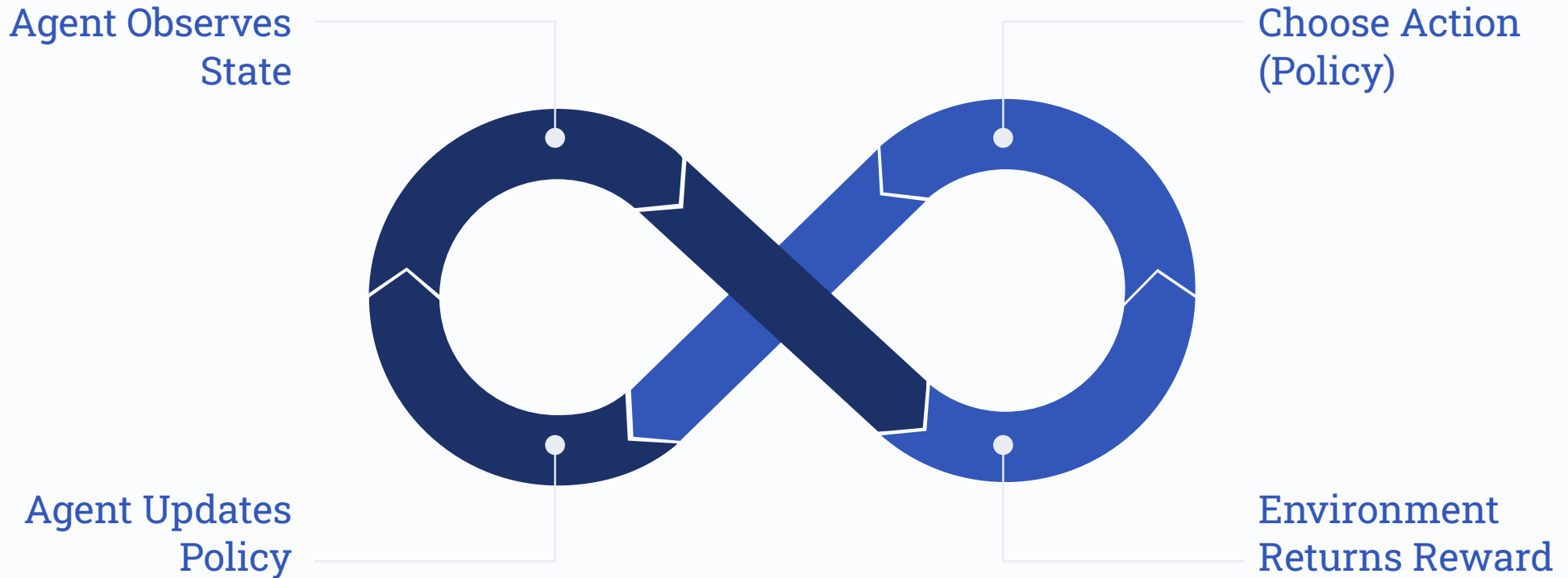
Reinforcement Learning is when a machine learns by **trial and error**, receiving rewards for good actions and penalties for bad ones. It's like training a pet with treats!



Reinforcement Learning - Key Terms

Term	What It Means	Video Game Analogy
Agent	The thing that's learning	You (the player)
Environment	The world the agent interacts with	The video game world
State	The current situation or observation	What you see on the screen right now
Action	Something the agent can do	Move left, jump, press button
Reward	Positive feedback for good actions	Getting points, coins, winning
Penalty	Negative feedback for bad actions	Losing a life, falling in a pit
Policy	The agent's strategy - what action to take in each state	Your game strategy - "when I see an enemy, I jump"
A/B Testing	Testing different strategies to see which gets better rewards	Trying different game strategies to see which gets more points

How It Works



The agent tries actions, receives feedback from the environment, updates its strategy, and repeats - thousands or millions of times, getting better and better!

Real-World Examples You Know

AlphaGo

Learned to play Go by playing millions of games against itself, getting rewards for winning

Self-Driving Cars

Learn to drive by getting rewards for staying on road, penalties for mistakes

Game AI

Learns to play video games by getting points for good moves

Robot Navigation

Learns to walk by getting rewards for moving forward, penalties for falling

- ❏ Reinforcement Learning is exactly like playing a video game: You start not knowing what to do. You try things - move left, jump, press buttons. You get points (rewards) when you do well. You lose points (penalties) when you make mistakes. Over time, you learn what actions lead to the most points. The machine does this thousands or millions of times, getting better and better!



Think This Way

Imagine you're learning to bake cookies 🍪



Supervised

Someone gives you a recipe with exact measurements and shows you examples of perfect cookies



Unsupervised

You look at many cookies and notice patterns - round ones are usually sweet, flat ones are crispy



Reinforcement

You try different recipes; when cookies taste good, you feel rewarded and use that recipe again

Mini Activity 2 : Identify the Learning Type

For each real-world application, identify which type of machine learning it uses (or might use):

#	Application	ML Type(s)	Why?
1	Netflix recommending shows		
2	A self-driving car staying in its lane		
3	Google Photos grouping pictures of the same person		
4	Your phone's face unlock feature		
5	AlphaGo beating world champions at Go		
6	Amazon suggesting "customers who bought this also bought..."		
7	Gmail's spam filter		
8	A robot learning to walk after many falls		
9	Spotify's Discover Weekly playlist		
10	A bank detecting fraudulent credit card transactions		

Mini Activity 2 : Answers

#	Application	ML Type(s)	Why?
1	Netflix recommendations	Unsupervised + Supervised	Finds patterns (unsupervised) + learns from ratings (supervised)
2	Self-driving car staying in lane	Reinforcement	Learns from rewards (staying safe) and penalties (going off road); observes state (lane position)
3	Google Photos grouping	Unsupervised (clustering)	Finds patterns without labels
4	Face unlock	Supervised (classification)	Trained on labeled photos of your face
5	AlphaGo	Reinforcement	Learned by playing millions of games and getting rewards; state = board positions
6	Amazon suggestions	Unsupervised (clustering)	Finds patterns in purchase data
7	Gmail spam filter	Supervised (classification)	Trained on labeled emails (spam/not spam)
8	Robot learning to walk	Reinforcement	Trial and error with rewards/penalties; state = position, balance
9	Discover Weekly	Unsupervised + Supervised	Finds patterns + learns from your listening
10	Fraud detection	Unsupervised	Finds unusual patterns that might be fraud

Discuss: Notice how some applications use more than one type! That's common in real AI systems.

Comparison - How They're Different

Let's put all three types side by side to see the differences clearly.

Aspect	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data Type	Labeled data (with answers)	Unlabeled data (no answers)	No data at start - learns from states, actions, rewards
Main Tasks	Classification, Regression	Clustering	Learning strategies
Goal	Predict correct label or value	Find hidden patterns	Maximize rewards over time
Feedback	Correct answers during training	No feedback - just patterns	Rewards and penalties after actions
Analogy	Studying with flashcards	Exploring and organizing	Playing a video game



Real AI Systems Combine Multiple Types

Here's an important insight: **Real-world AI systems often use multiple types of machine learning together.** They're not limited to just one approach!

Combined ML Types

Example : Netflix Recommendations

Netflix doesn't use just one type - it combines several:



Collaborative Filtering

Unsupervised - Finds patterns: "People who watched this also watched that"



Personalization

Supervised - Learns from your ratings (labeled data about your preferences)



Content Analysis

Supervised - Analyzes movie descriptions, genres, actors



A/B Testing

Reinforcement - Tests different recommendation strategies
"Try A, try B - keep what works best."

More Real-World Examples

Self-Driving Cars

- **Object Detection (Supervised):** Recognizes pedestrians, signs, other cars (trained on labeled images)
- **Path Planning (Reinforcement):** Learns optimal routes and driving strategies
- **Traffic Pattern Analysis (Unsupervised):** Finds patterns in traffic flow at different times

Voice Assistants (Siri, Alexa)

- **Speech Recognition (Supervised):** Converts sound to text (trained on labeled audio)
- **Understanding Intent (Supervised):** Figures out what you want (trained on labeled requests)
- **Personalization (Unsupervised):** Learns your habits, favorite music, common requests
- **Response Generation (Reinforcement):** Tests different responses, learns which work best

Summary : What we Learned Today

Supervised Learning

Machine learning with labeled examples - like studying with flashcards that have answers on the back

Classification

Supervised task where AI predicts a category (spam/not spam, cat/dog)

Regression

Supervised task where AI predicts a number (temperature, price)

Unsupervised Learning

Machine learning that finds patterns without labels - like organizing a messy closet without instructions

Clustering

Unsupervised task where AI finds hidden groups in data

Reinforcement Learning

Machine learning through trial and error with rewards and penalties - like playing a video game

Your Homework: ML Type Hunter

Part 1: Quick Warm-Up

Match each learning type to its description:

Learning Type	Description
1. Supervised Learning	A. Learning from rewards and penalties
2. Unsupervised Learning	B. Learning with labeled examples
3. Reinforcement Learning	C. Finding patterns without labels

Answers: 1-___ , 2-___ , 3-___

Part 2: Your Mission - Find ML Types

Find one real example of each type of machine learning. They can be apps, websites, or devices you use.

ML Type	Example (App/System)	What does it do?	What makes it this type?
Supervised			
Unsupervised			
Reinforcement			

Need ideas?

- **Supervised:** Face unlock, spam filter, handwriting recognition, Google Translate
- **Unsupervised:** Google News, Amazon "customers also bought," Spotify recommendations, YouTube suggestions
- **Reinforcement:** Game AI (chess, Go), self-driving cars, robot vacuum learning your floor plan, AlphaGo

Part 3: Challenge - Find a System That Combines Types

Can you find an example of a real AI system that probably uses more than one type of machine learning?

- System: ____
- Supervised part: ____
- Unsupervised part: ____
- Reinforcement part: ____

Example (Netflix):

- **Supervised:** Learns from your ratings (labeled data)
- **Unsupervised:** Finds patterns in what people watch together
- **Reinforcement:** Tests different recommendations, keeps what works